Ethanol Blending in Gasoline – Estonia

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U.S. GRAINS & BIOPRODUCTS COUNCIL

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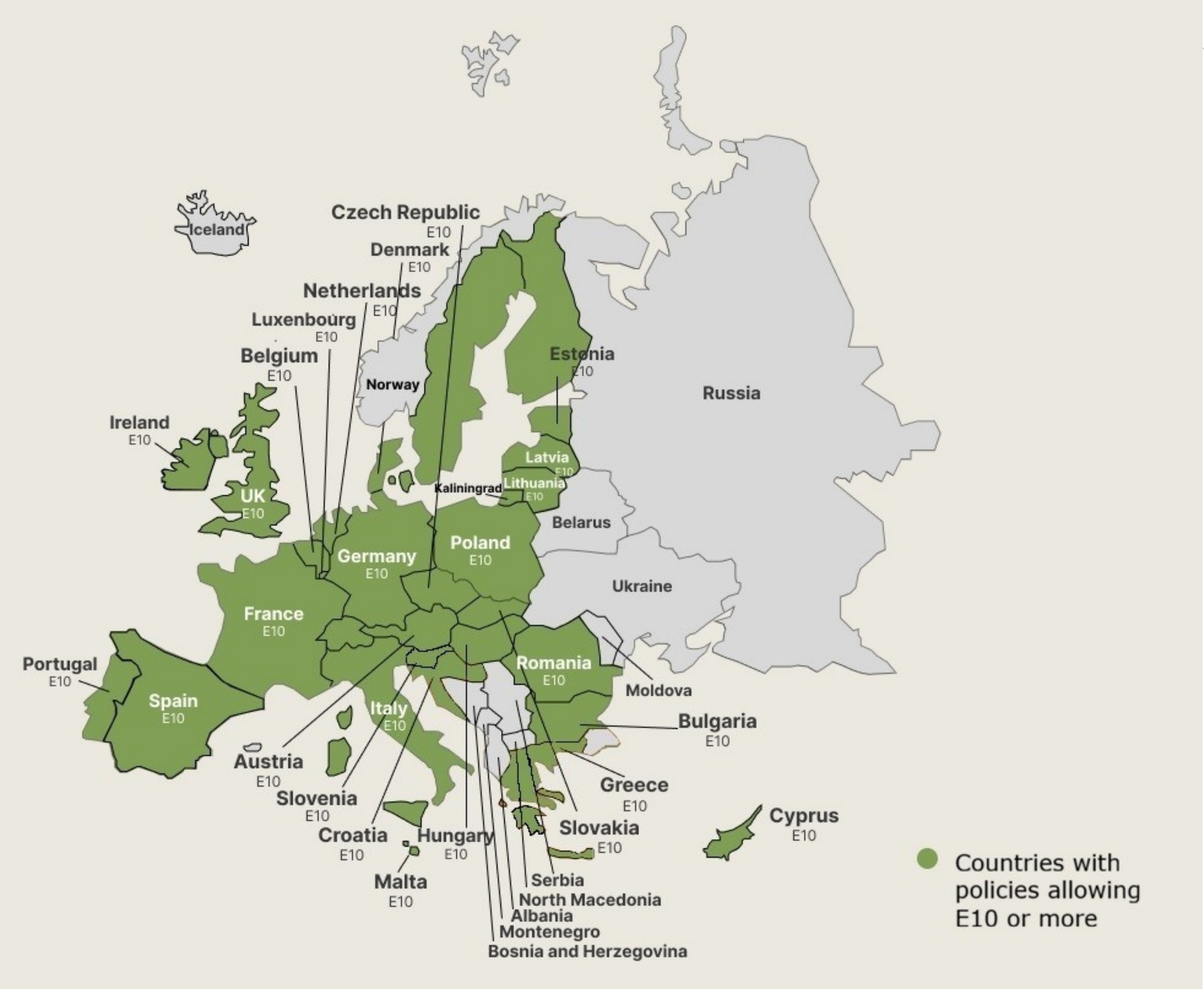
Ethanol Blending in Europe

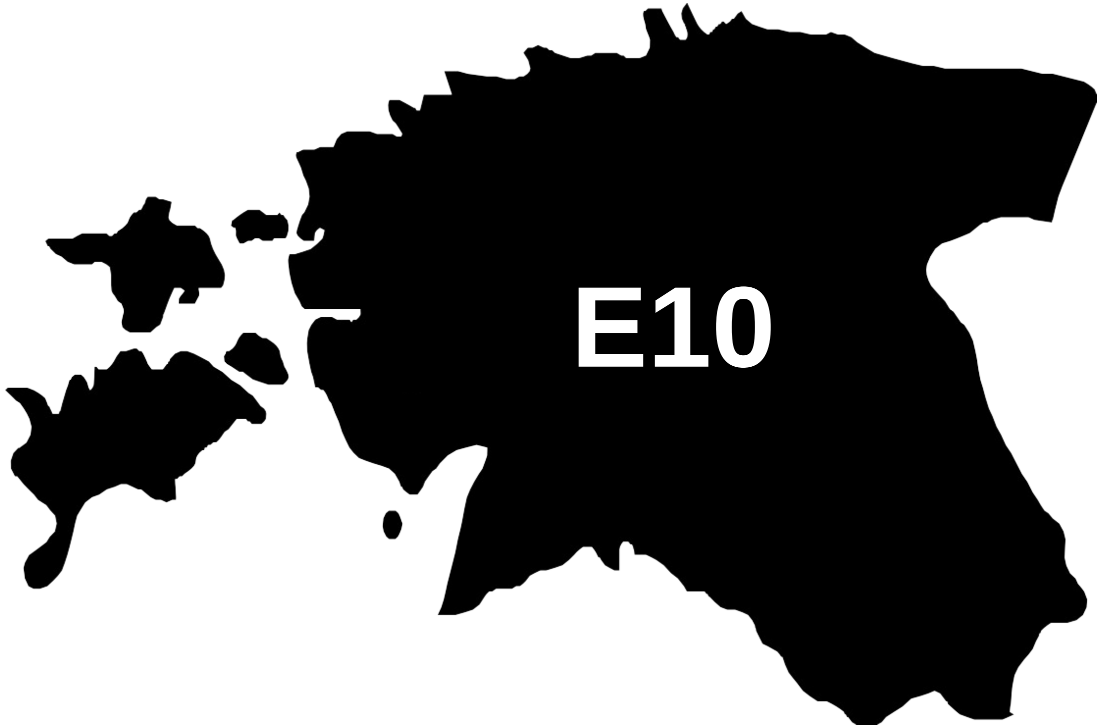
There are important fuel quality and environmental impact of vehicle emission challenges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Europe to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.



Twenty eight countries with potential and additional use of ethanol were studied: 1) Gasoline market profiles; 2) Optimization of gasoline blends with ethanol and, 3) Environmental impact of gasolines blended with ethanol.

Ethanol Gasoline Blending in Estonia

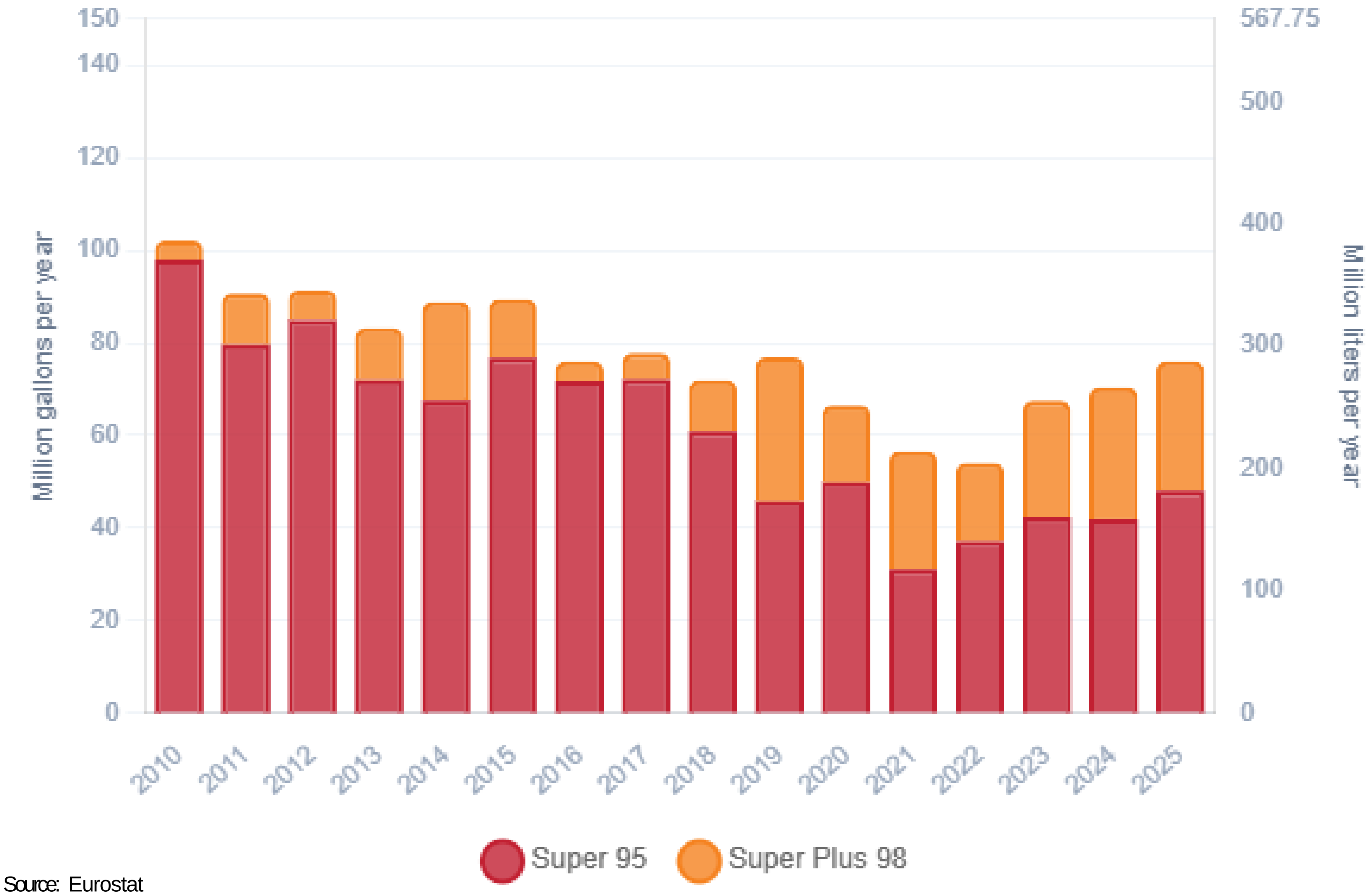


In 2025, gasoline consumption in Estonia was 286 million liters (76 million gallons) and growth rate was -1.9%. There is no gasoline production. Gasoline net imports were 341 million liters (90 million gallons). Estonia complies with the European gasoline standards and offers two main grades: RON 95 and RON 98 with an average octane of 96.1 RON.

In 2025, ethanol consumption in Estonia was 5 million liters (1 million gallons) representing 2% of gasoline demand. Ethanol imports were 5 million liters (1 million gallons). Ethanol sources is Cereals 100%.

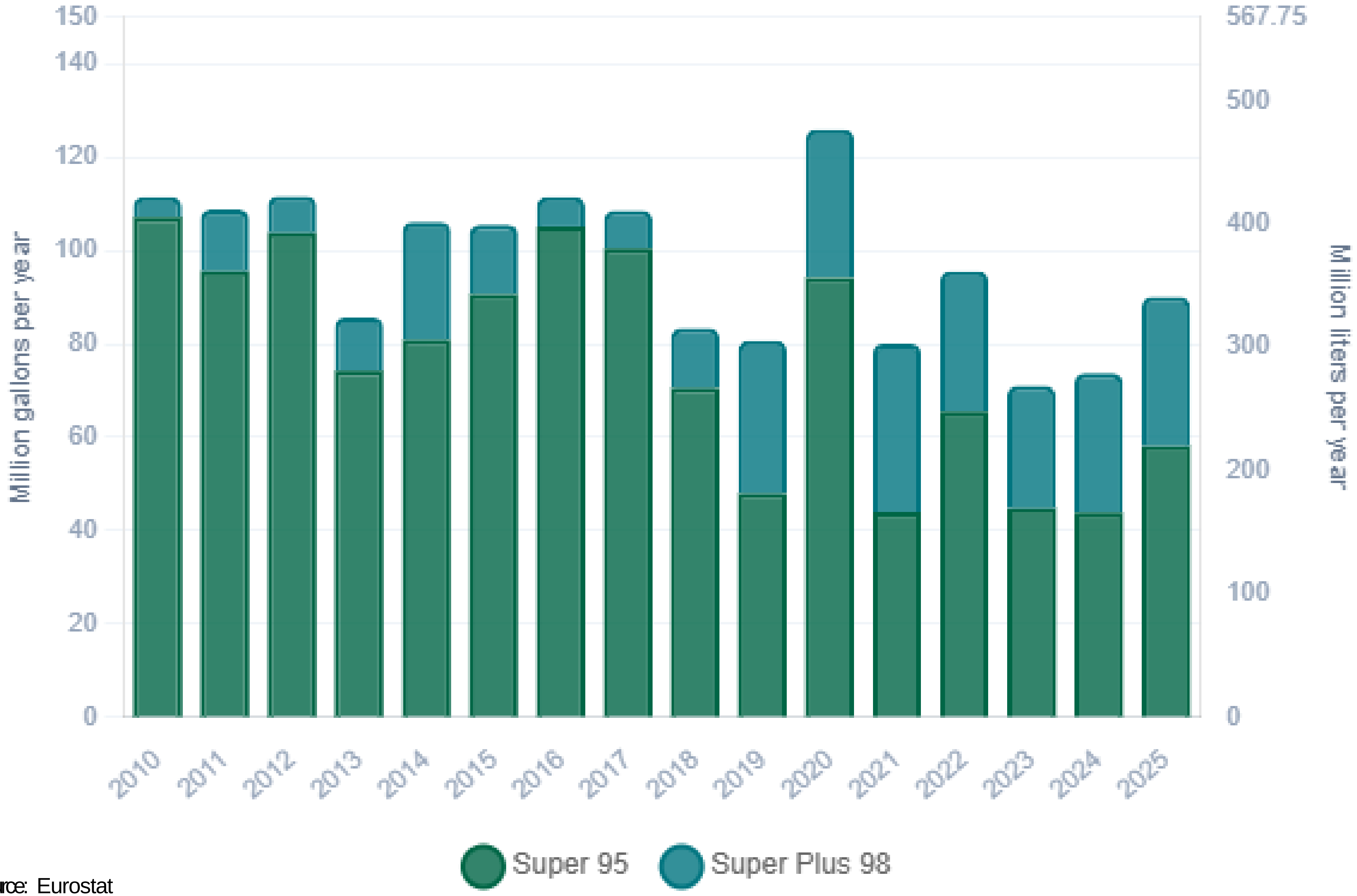
Source: Eurostat, Eurpean Commision

Gasoline Demand in Estonia



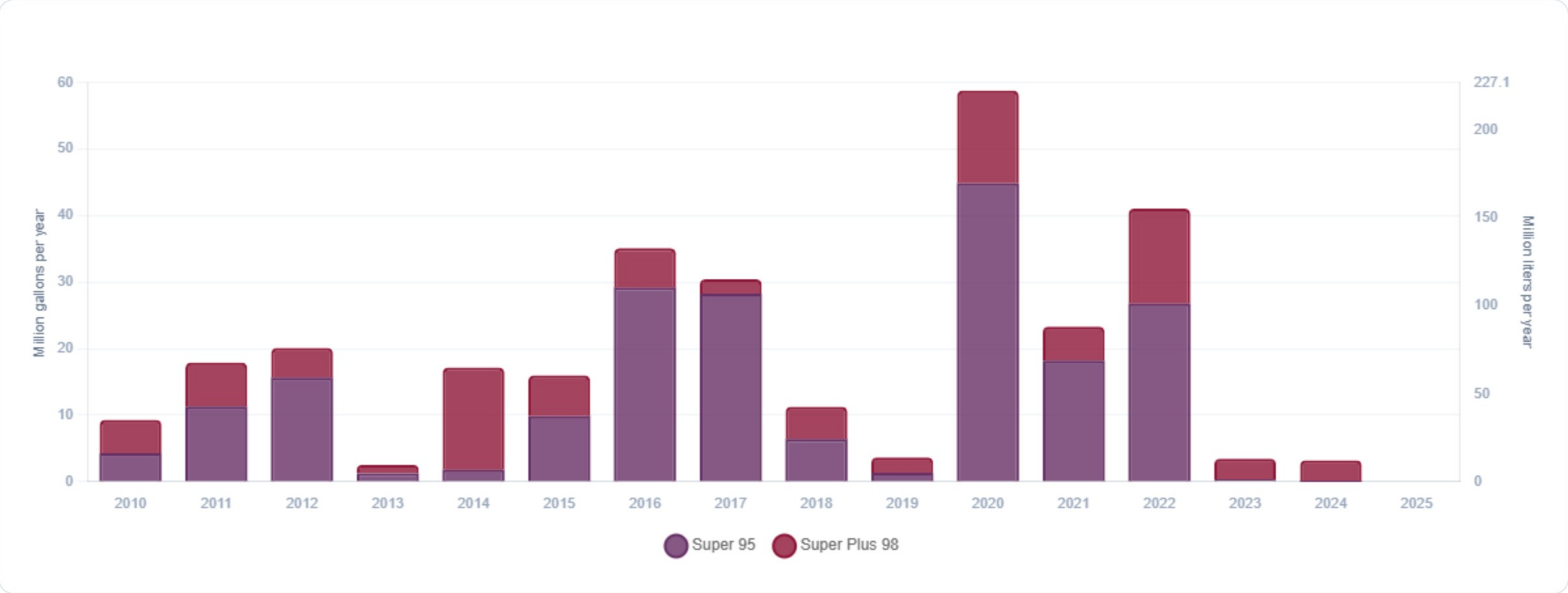
Source: Eurostat

Gasoline Imports in Estonia



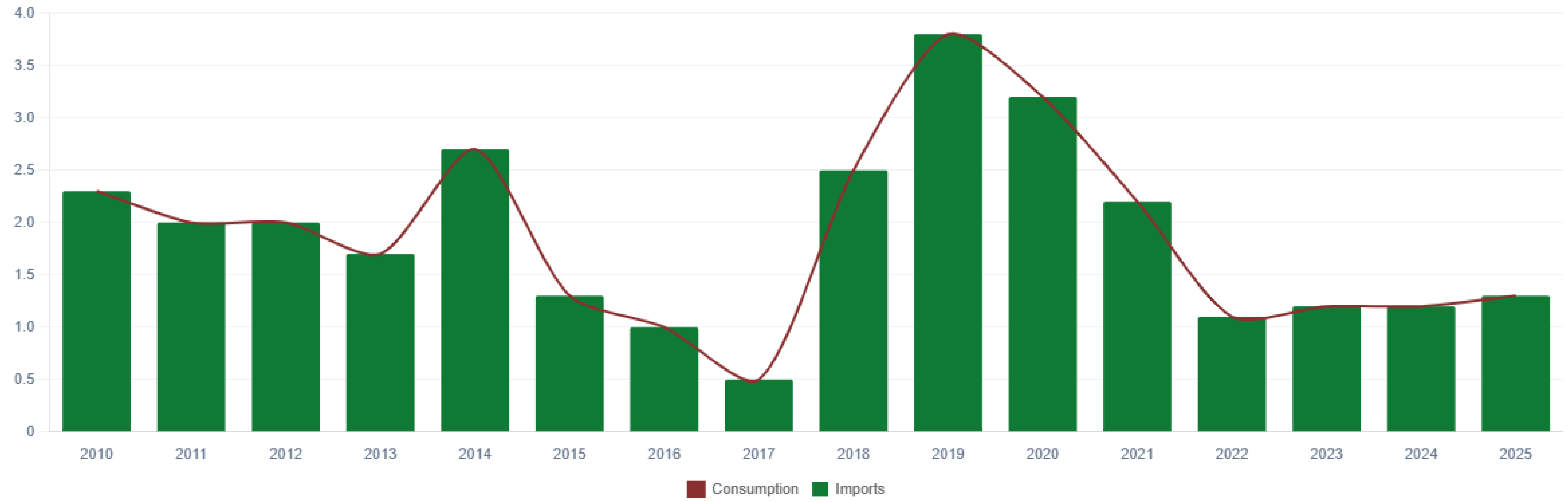
Source: Eurostat

Gasoline Exports in Estonia



Source: Eurostat

Ethanol Balance in Estonia



Source: Eurostat, Statistics Estonia

Gasoline Quality in Estonia

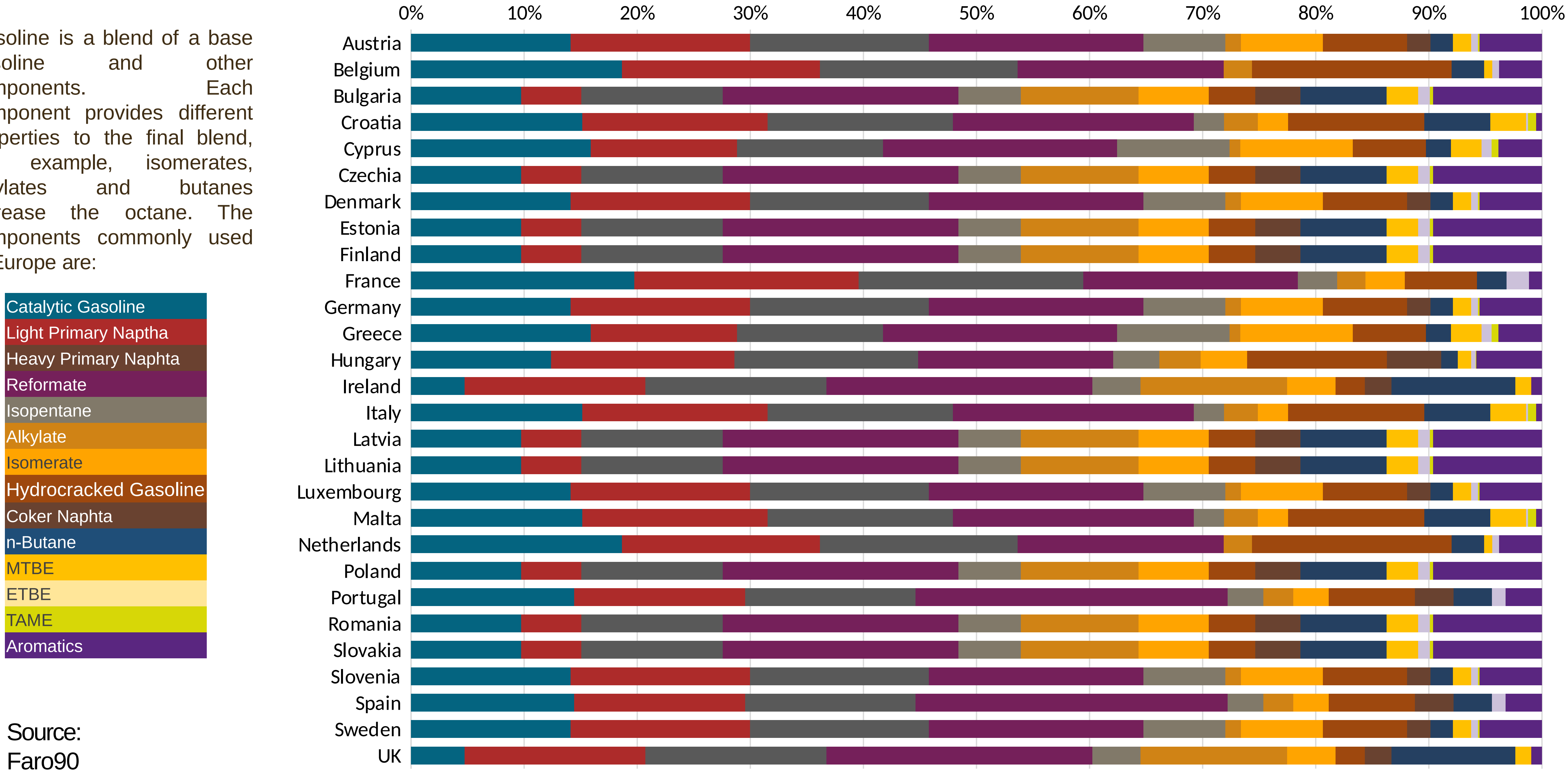
Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	7.5							
Advanced Biofuels (%cal)	0.5							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	-							

* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commision

Gasoline Component Blending in Europe

Gasoline is a blend of a base gasoline and other components. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Europe are:



Source: Faro90

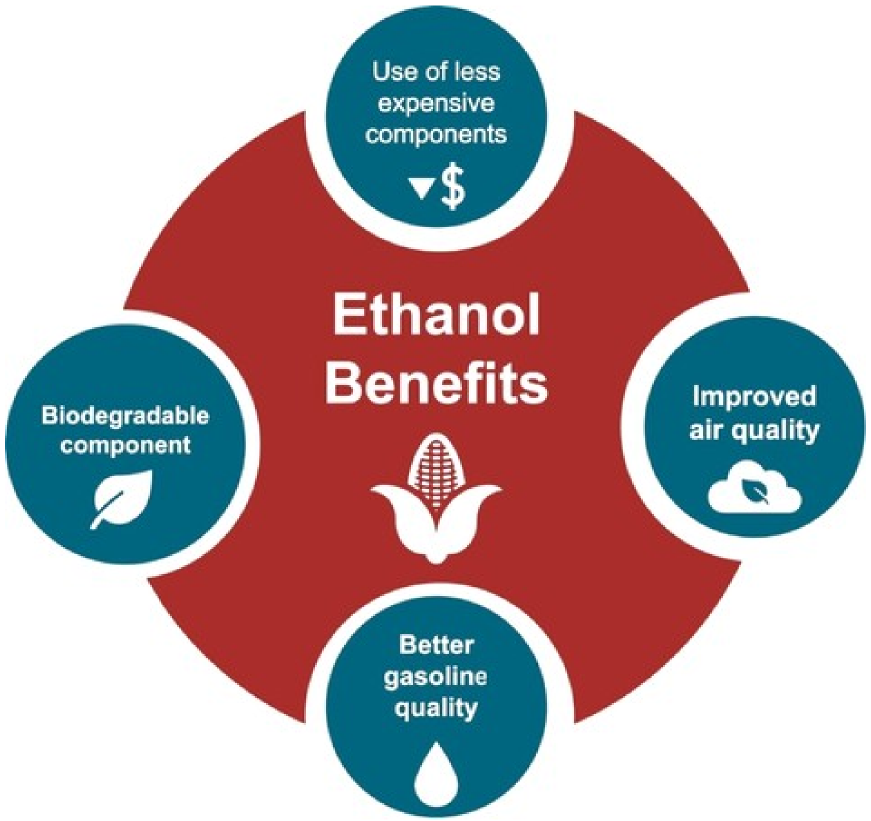
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component.The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives.To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

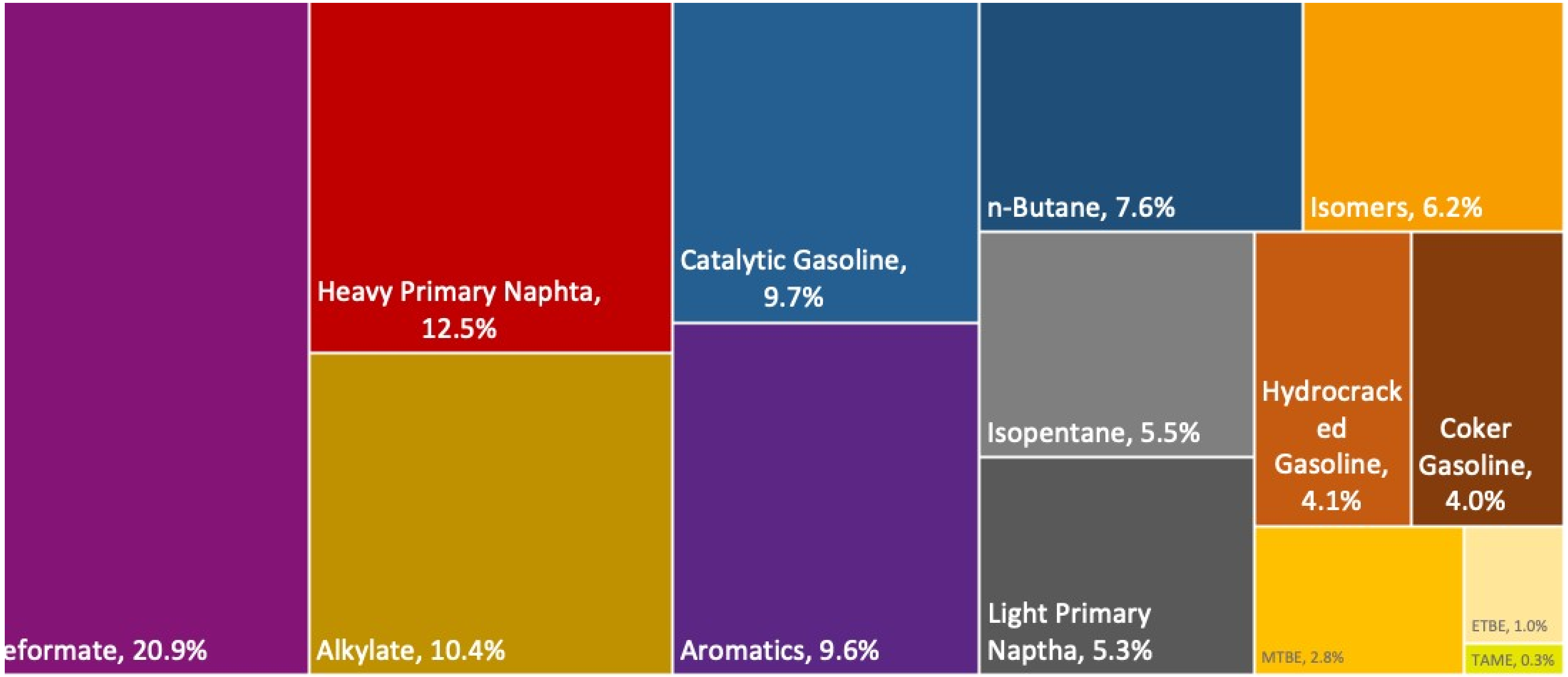
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

The blending model uses gasoline component spot average prices 2025 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.

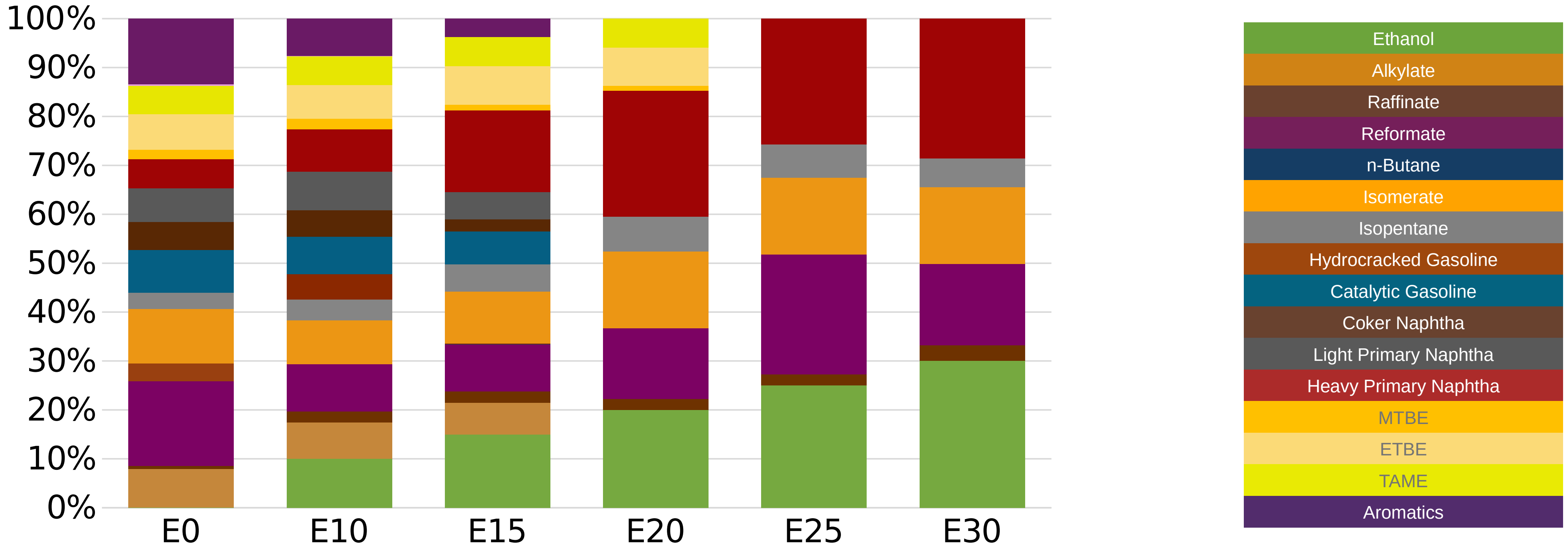


Estonia Available Blending Components



Source: Faro90

Estonia – Gasoline 95 – Constant Octane

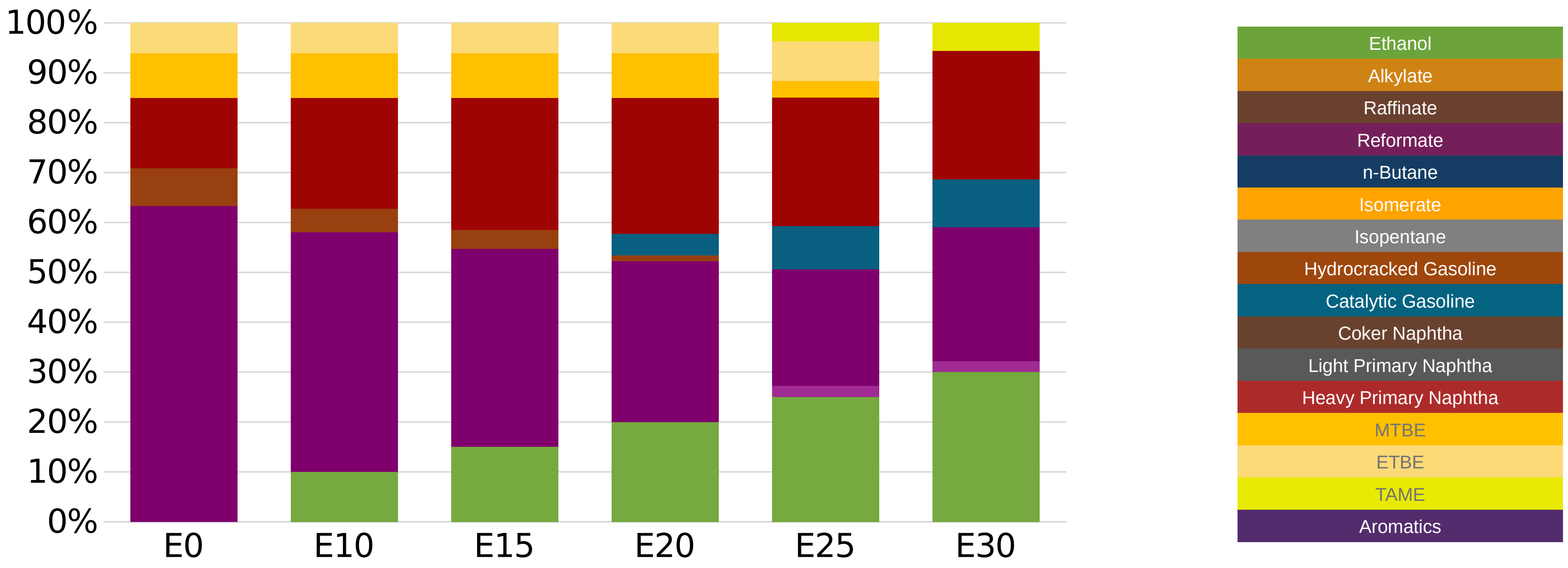


Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.8
Price (US$/gal)	$ 2.071	$ 1.933	$ 1.833	$ 1.718	$ 1.672	$ 1.619

Average CIF 2025 Prices

Source: Faro90

Estonia - Gasoline 98 - Constant Octane

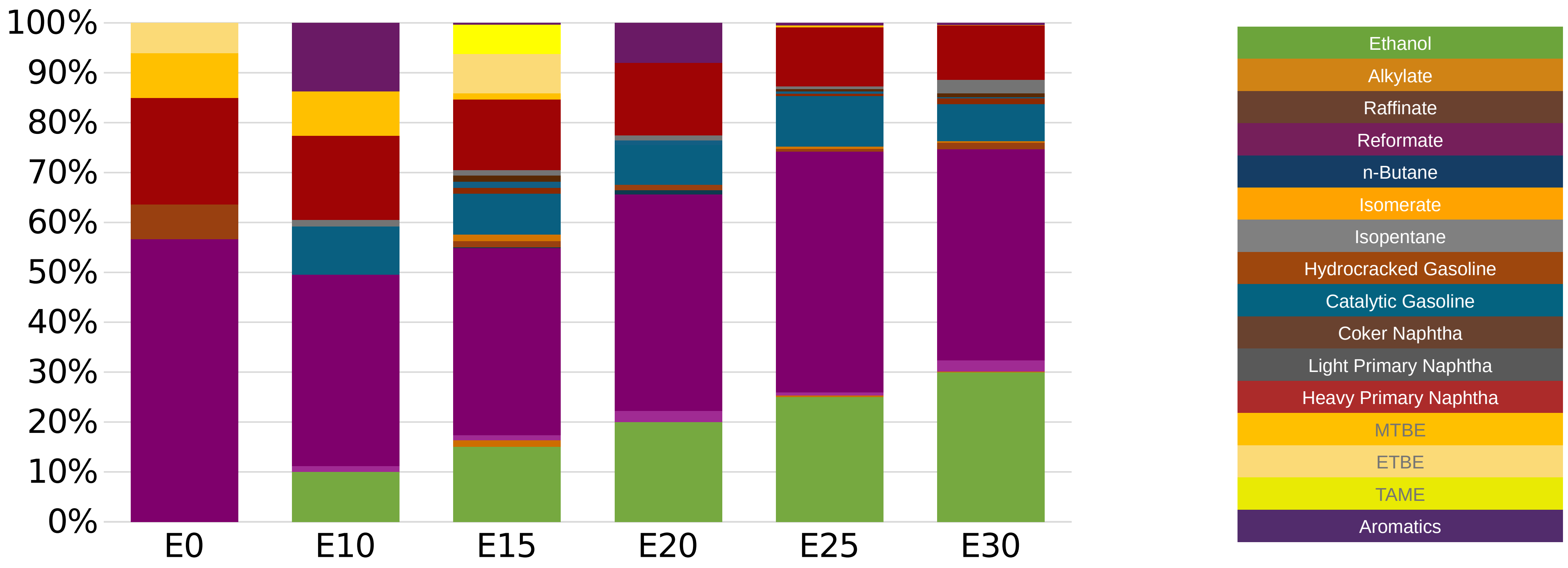


Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US$/gal)	$ 2.091	$ 1.974	$ 1.905	$ 1.833	$ 1.768	$ 1.718

Average CIF 2025 Prices

Source: Faro90

Estonia – Gasoline 95 – Increased Octane

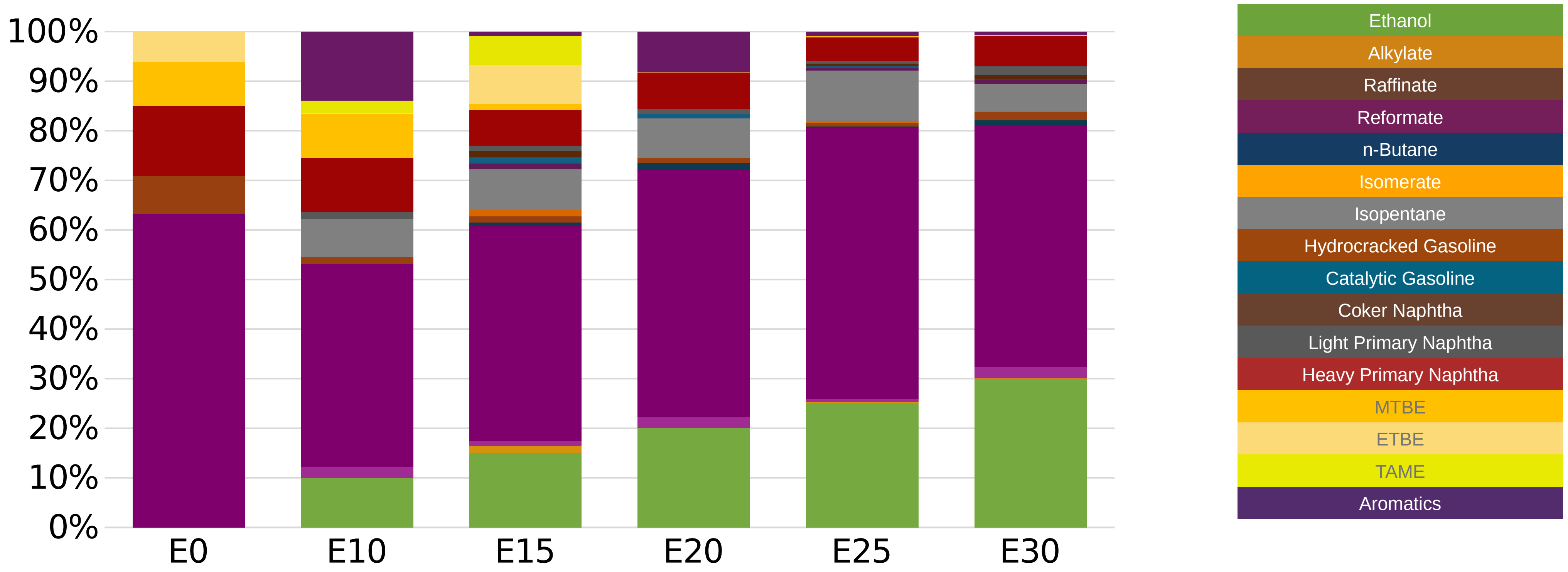


Octane (RON)	95.0	96.2	98.3	98.7	100.5	101.8
Price (US$/gal)	$ 1.959	$ 1.959	$ 1.959	$ 1.959	$ 1.959	$ 1.959

Average CIF 2025 Prices

Source: Faro90

Estonia – Gasoline 98 – Increased Octane



Octane (RON)	98.0	99.0	100.9	101.2	102.8	104.1
Price (US$/gal)	$ 2.091	$ 2.091	$ 2.091	$ 2.091	$ 2.091	$ 2.091

Average CIF 2025 Prices

Source: Faro90

Vehicle Emission Impact for Ethanol Gasoline Blending

The model used in this analysis takes as a reference the **International Vehicle Emissions Model (IVE).**

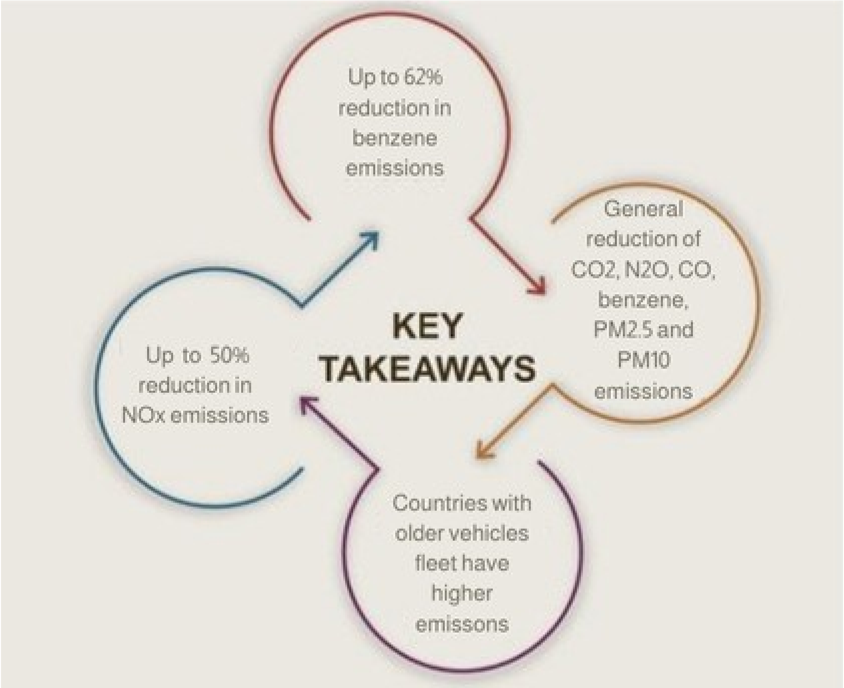
The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

Main Results



**Source: Bureau of transportation statistics.*

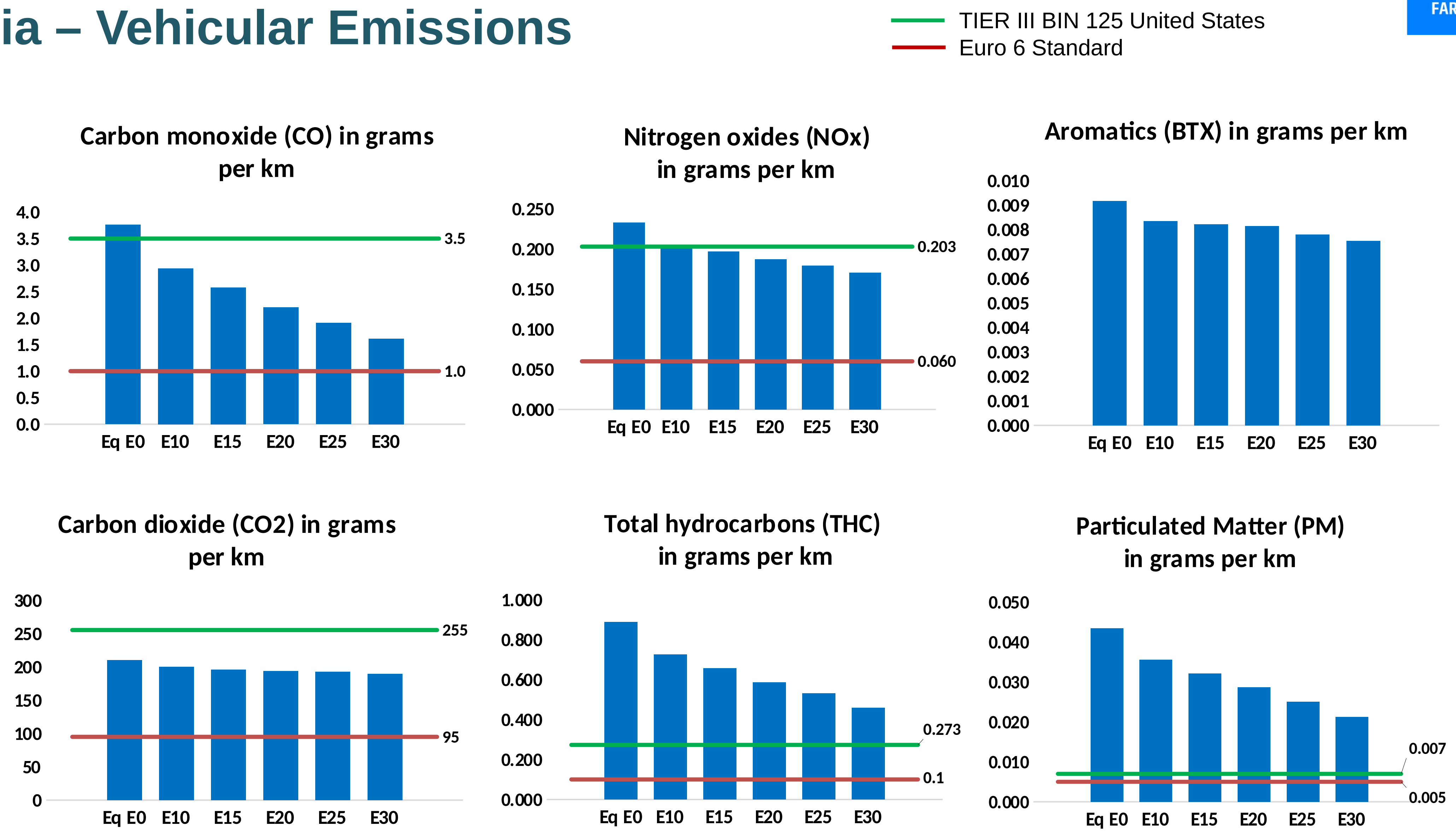
Gasoline Vehicle Fleet - Estonia

Type	Weight	Engine	Exhaust	Recovery	Age	No. Vehicles
Auto/Sml Truck	Light	Carburator	3-Ways	PCV Tank	>8 years	202
		Single FI	3-Ways		>8 years	6,340
		Multi FI	3-Ways		>8 years	64,777
			Euro III		>8 years	85,307
			Euro IV		>8 years	186,431
			Euro V		>8 years	70,437
			Euro 6		>8 years	168,893
					4-8 years	120,545
					<4 years	20,446
			Hybrid		All	12,297
		Electric			All	12,297
	Medium	Carburator	3-Ways	PCV Tank	>8 years	20
		Single FI	3-Ways		>8 years	169
		Multi FI	3-Ways		>8 years	9,814
			Euro III		>8 years	17,916
			Euro IV		>8 years	65,469
			Euro V		>8 years	21,688
			Euro 6		>8 years	58,575
					4-8 years	64,624
					<4 years	16,669
			Hybrid		All	63,148
		Electric			All	4,334
Sml Engines	Light	4-Cycle	None	PCV	>8 years	91
			Euro III		>8 years	186
			Euro IV		>8 years	2,376
			Euro V		>8 years	952
			Euro 6		>8 years	4,061
					4-8 years	4,969
					<4 years	7,086
			Hybrid		All	1,521
		Electric			All	335
Truck	Heavy	Carburator	3-Ways	PCV	>8 years	1,113
		Fuel-Injection	3-Ways		>8 years	8,013
			Euro III		>8 years	4,753
			Euro IV		>8 years	23,826
			Euro V		>8 years	8,719
			Euro 6		>8 years	31,501
					4-8 years	39,741
					<4 years	31,547
			Hybrid		All	100
		Electric			All	623

Vehicle Fleet: 1,241,909
Gasoline Vehicle Fleet: 589,169

Source: Eurostat, ACEA

Estonia – Vehicular Emissions



Source: Faro90

Estonia – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	56,434	50,255	44,075	38,652	33,029	28,656	24,190	-22%	-41%
CO2	3,158,152	3,076,942	3,000,244	2,939,923	2,910,074	2,892,954	2,839,628	-5%	-8%
VOC	13,317	12,098	10,880	9,843	8,783	7,962	6,885	-18%	-34%
NOx	3,497	3,229	3,048	2,956	2,808	2,693	2,556	-13%	-20%
SOx	37	34	31	29	26	24	21	-15%	-28%
NH3	1,220	1,208	1,196	1,201	1,215	1,224	1,232	-2%	0%
N2O	43	42	42	42	43	44	44	-1%	2%
CH4	2,643	2,643	2,643	2,683	2,741	2,791	2,839	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	55	52	49	47	45	44	41	-11%	-18%
Acetaldehyde	204	234	343	523	712	814	962	68%	249%
Formaldehyde	785	763	889	1,044	1,094	1,210	1,318	13%	39%
Benzene	138	132	125	123	122	117	113	-9%	-11%
PM2.5	187	170	153	138	123	108	91	-18%	-34%
PM10	466	423	381	344	307	268	227	-18%	-34%

Source: Faro90

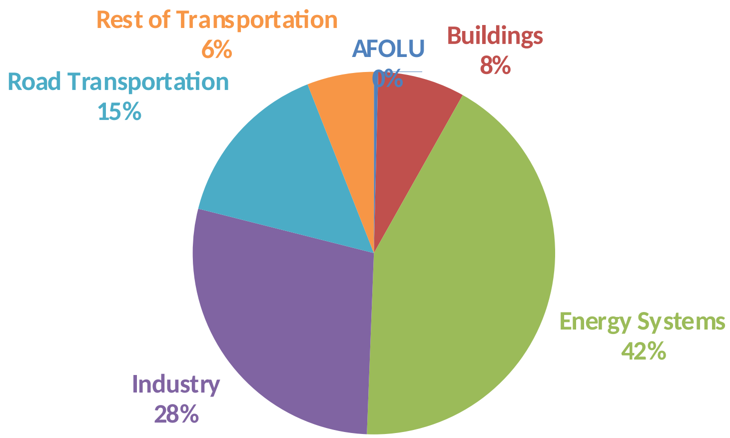
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

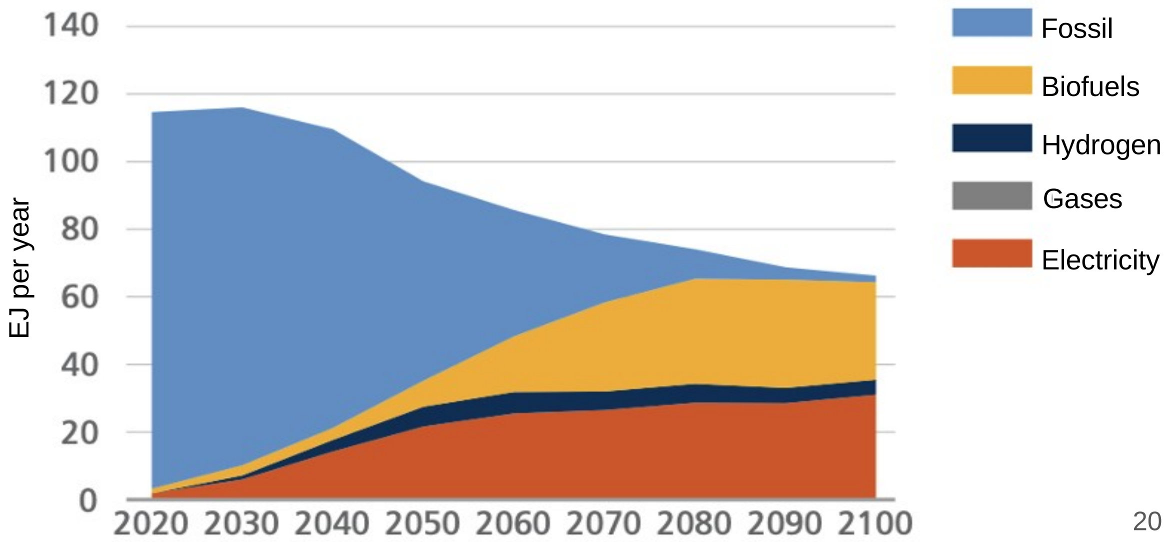
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE).** Results are compared to the current country targets for 2035. Current 2025 country emission are those published by **IPCC**.

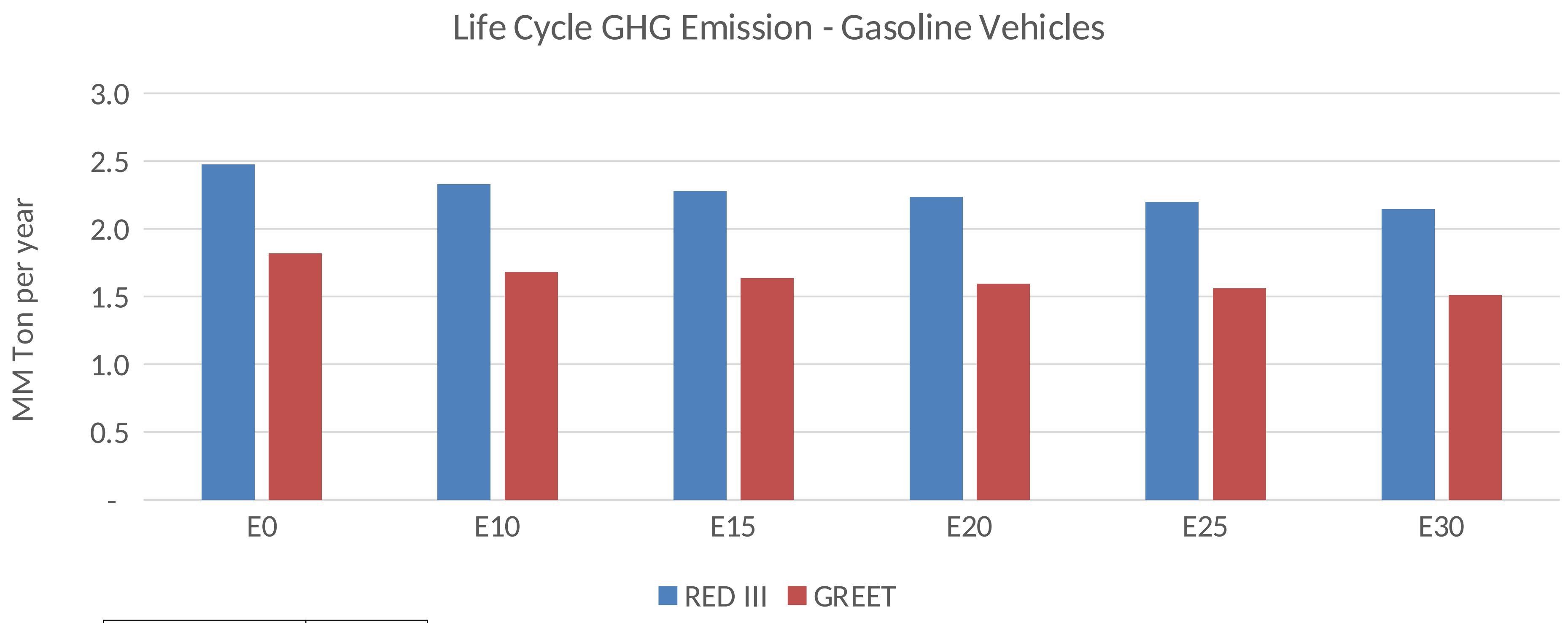
Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



Estonia – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2025 (MMT/año)	15.6
Target @ 2035 (MMT/year)	8.8
Delta	6.8

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90